

Data Mining Numerical Problems & Solutions

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · Chapter 2: Data Preprocessing

Ch 2 – Data Preprocessing – Numerical Problems & Solutions

7 distinct numerical methods · 8 PYQ instances solved · 2 extra practice problems. Every answer independently recomputed, not copied from the source notes.

Methods in this chapter

- **M1** Normalisation: min-max, z-score, decimal scaling 78 Bh
- **M2** PCA: principal components + proportion of variance 76 Ch
- **M3** Distance measures + distance matrix (Euclidean, Manhattan, Supremum) 80 Bh, 76 Ch
- **M4** Cosine similarity 80 Bh, 80 Ba, 76 Ch
- **M5** Binary similarity: Jaccard, SMC, symmetric dissimilarity 80 Ba, 76 Ch
- **M6** Schema design + OLAP operation sequence 78 Bh, 76 Ch
- **M7** Central tendency, quartiles, IQR and outliers **NEW**, no PYQ

Every problem below ends with its solution drawn step by step. Each panel carries a chip naming the move it makes; the decisive step is framed **KEY STEP**. The table says what each move is and where it is used.

The moves every chapter-2 numerical below is built from

Move	What it says (where: problem, step; orange = key step)	Move	What it says (where: problem, step; orange = key step)
map the range	Min-max: slide min to new_min and stretch max to new_max; order and gaps kept. 1.1 1; 1.2 2; Practice B 1	dot product	Multiply component by component and add. 4.1 1; 4.2 1; 4.3 1
centre on mean	Z-score: subtract the mean, divide by σ ; the result counts standard deviations. 1.1 2; 1.2 2; 7.4 1	length	$\ d\ = \sqrt{\sum d_k^2}$ for each vector. 4.1 2; 4.2 2; 4.3 2
pick j from max	Decimal scaling: the smallest j with $\max v /10^j$ STRICTLY below 1. 1.1 3; 1.2 2	divide	$\cos = \text{dot} / (\text{product of lengths})$; the angle, magnitude cancelled. 4.1 3; 4.2 3; 4.3 3
equal-depth bins	Sort, then cut into bins holding the same COUNT (depth), not the same range. 1.2 1; Practice A 1	tally the pairs	Count $M_{11}, M_{10}, M_{01}, M_{00}$ position by position. 5.1 1; 5.2 2,3,4
bin mean	Replace every value by its bin's mean. 1.2 1; Practice A 1	split attributes	Decide which attributes are asymmetric (a shared 0 means nothing) and which symmetric. 5.2 1
bin boundary	Replace every value by the nearer of its bin's min and max. Practice A 2	Jaccard	$M_{11}/(M_{11} + M_{10} + M_{01})$: 0-0 matches ignored. 5.1 2; 5.2 2
equal-width bins	Cut the range into equal intervals: $W = (\max - \min)/N$; counts differ. Practice A 3	SMC	$(M_{11} + M_{00})/\text{all}$: 0-0 matches count. 5.1 3; 5.2 4
block split	A row and column of zeros decouple: solve each block alone. 2.1 1	symmetric d	$(M_{10} + M_{01})/\text{all} = 1 - \text{SMC}$. 5.2 3
characteristic eq.	$ \Sigma - \lambda I = 0$ gives the eigenvalues. 2.1 2	roll-up	Climb a dimension's hierarchy to a coarser level. 6.1 1,2,3; 6.2 1,2
trace check	The eigenvalues must add up to the trace (sum of the diagonal). 2.1 3	roll-up to all	A dimension the question never mentions is summed away. 6.1 4; 6.2 3
eigenvector	Solve $(\Sigma - \lambda I)v = 0$ from one row, then scale to unit length. 2.1 4	dice	Restrict two or more dimensions at once. 6.1 5
share of variance	$\lambda_i / \sum \lambda$: how much of the spread each component keeps. 2.1 5	slice	Restrict exactly one dimension. 6.2 4
differences	Subtract attribute by attribute first; every distance is built from these. 3.1 1; 3.2 1	cumulative freq.	Run the frequencies up to find the interval holding the $N/2$ -th value. 7.1 1; 7.2 1
Euclidean	$\sqrt{\sum d_k^2}$: the straight line ($r = 2$). Warm-up 1; 3.2 2	interpolate	Assume values spread evenly inside the interval and walk in proportionally. 7.1 2; 7.2 2
Manhattan	$\sum d_k $: along the grid ($r = 1$). Warm-up 2; 3.1 2	empirical rule	$\text{mean} - \text{mode} \approx 3(\text{mean} - \text{median})$. 7.3 1
supremum	$\max d_k $: the single largest difference ($r \rightarrow \infty$). Warm-up 3; 3.1 3	quartile position	Q_k sits at position $k(n+1)/4$; interpolate between neighbours. 7.4 2
fill by symmetry	A distance matrix has a zero diagonal and is symmetric: compute the lower triangle only. 3.1 2,3	1.5 IQR fences	Outside $Q_1 - 1.5 IQR$ or $Q_3 + 1.5 IQR$ is a suspected outlier. 7.4 3

M1. Normalisation

Method

- Min-max maps v into $[new_min, new_max]$, preserves relationships exactly:

$$v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

- Z-score uses mean $\bar{A}$ and std deviation σ_A . Use when min/max unknown, or outliers would squash a min-max result:

$$v' = \frac{v - \bar{A}}{\sigma_A} \quad \sigma_A = \sqrt{\frac{1}{n} \sum (v_i - \bar{A})^2}$$

- Decimal scaling moves the decimal point:

$$v' = \frac{v}{10^j}, \quad j = \text{smallest int such that } \max(|v'|) < 1$$

Two traps. (1) Z-score uses the **mean**, not the min. (2) For decimal scaling, j must make the max *strictly* less than 1, so $\max |v| = 1000$ needs $j = 4$, not $j = 3$.

Problem 1.1 2+2+2

QUESTION AS PRINTED 2078 Bhadra · Q3 · [2+2+2]

Use the following methods to normalize the data: 200, 300, 400, 600 and 1000.

- Min-max normalization by setting min = 0 and max = 1
- Z-score normalization
- Normalization by decimal scaling

Given

$v = \{200, 300, 400, 600, 1000\}$, $n = 5$, $min_A = 200$, $max_A = 1000$.

(a) Min-max, $new_min = 0$, $new_max = 1$

$$v' = \frac{v - 200}{1000 - 200} = \frac{v - 200}{800}$$

v	v'
200	$0/800 = 0$
300	$100/800 = 0.125$
400	$200/800 = 0.25$
600	$400/800 = 0.5$
1000	$800/800 = 1$

(c) Decimal scaling

- $\max |v| = 1000$.
- $j = 3$: $1000/10^3 = 1$, not < 1 . **Rejected.**
- $j = 4$: $1000/10^4 = 0.1 < 1$. **Use $j = 4$.**

v	$v' = v/10000$
200	0.02
300	0.03
400	0.04
600	0.06
1000	0.1

Final answer

- Min-max: 0, 0.125, 0.25, 0.5, 1
- Z-score: -1.0607, -0.7071, -0.3536, +0.3536, +1.7678
- Decimal scaling: 0.02, 0.03, 0.04, 0.06, 0.1

(b) Z-score

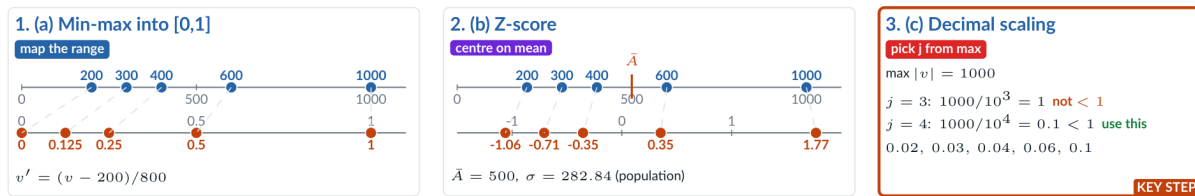
- Mean $\bar{A} = \frac{200 + 300 + 400 + 600 + 1000}{5} = \frac{2500}{5} = 500$
- Deviations: -300, -200, -100, +100, +500
- $\sum (v - \bar{A})^2 = 90000 + 40000 + 10000 + 10000 + 250000 = 400000$
- $\sigma_A = \sqrt{400000/5} = \sqrt{80000} = 282.84$

v	$v' = (v - 500)/282.84$
200	$-300/282.84 = -1.0607$
300	$-200/282.84 = -0.7071$
400	$-100/282.84 = -0.3536$
600	$+100/282.84 = +0.3536$
1000	$+500/282.84 = +1.7678$

Note on σ verified/added

- Above uses the **population** std deviation, $\sqrt{\sum (v - \bar{A})^2 / n}$. This is the form Han & Kamber use, so it is the safe default.
- If the examiner wants the **sample** std deviation, divide by $n - 1$: $s = \sqrt{400000/4} = 316.23$, giving -0.9487, -0.6325, -0.3162, +0.3162, +1.5811.
- State which one you used.** Either is accepted if labelled.

Step by step: three ways to normalise the same five values



Problem 1.2 – all four transformations on one data set 6

QUESTION AS PRINTED 2079 Baishakh · Q2 · [6]

Suppose that the data for analysis include the attribute the frequency of stop words in documents. The values are given in increasing order: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

- Use smoothing by bin means with a depth of 3.
- Use min-max normalization to transform the value 35 into the range from 0.0 to 1.0.
- Use z-score normalization to transform the value 35 where the standard deviation of the above frequency is 12.94.
- Use normalization by decimal scaling to transform the value 35.

$n = 27$, already sorted. “Depth 3” means 3 values per bin, not 3 bins – so $27/3 = 9$ bins. This is the single most common error on this question. Parts (b)–(d) each transform only the value 35, so no full table is needed there.

(a) Smoothing by bin means, depth 3

partition in order, replace every value by its bin mean

Bin	Values	Mean	Smoothed to
1	13, 15, 16	$44/3 = 14.67$	14.67, 14.67, 14.67
2	16, 19, 20	$55/3 = 18.33$	18.33, 18.33, 18.33
3	20, 21, 22	$63/3 = 21.00$	21, 21, 21
4	22, 25, 25	$72/3 = 24.00$	24, 24, 24
5	25, 25, 30	$80/3 = 26.67$	26.67, 26.67, 26.67
6	33, 33, 35	$101/3 = 33.67$	33.67, 33.67, 33.67
7	35, 35, 35	$105/3 = 35.00$	35, 35, 35
8	36, 40, 45	$121/3 = 40.33$	40.33, 40.33, 40.33
9	46, 52, 70	$168/3 = 56.00$	56, 56, 56

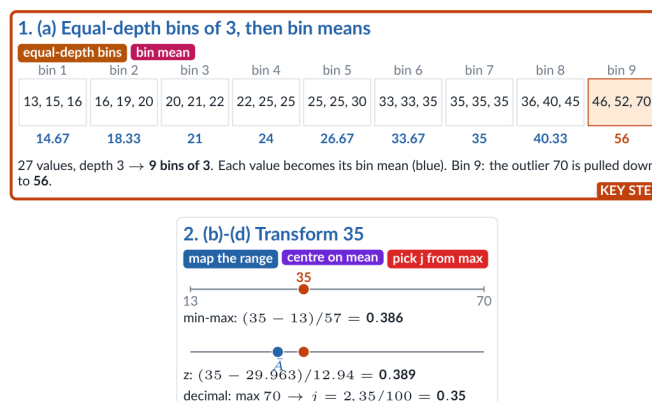
- Bin 9 shows why smoothing works: the outlier 70 is pulled down to 56, and 46 pushed up. Noise is absorbed into the local average.
- This is **equal-depth** (equal-frequency) binning: same count per bin, not the same value range. Compare Practice A, which uses depth 4.

(b), (c), (d): transforming the single value 35

Method	Formula	Substitution	Result
Min-max into [0, 1]	$v' = \frac{v - \min}{\max - \min} (new_{\max} - new_{\min}) + new_{\min}$	$\frac{35 - 13}{70 - 13} = \frac{22}{57}$	0.386
Z-score	$v' = \frac{v - \bar{A}}{\sigma_A}, \bar{A} = \frac{809}{27} = 29.963$	$\frac{35 - 29.963}{12.94} = \frac{5.037}{12.94}$	0.389
Decimal scaling	$v' = v/10^j$, smallest j with $\max(v') < 1$	$\max = 70 \rightarrow j = 2, 35/100$	0.35

- σ is given as 12.94, so do not recompute it – but you *must* compute the mean yourself. Sum the 27 values: 809.
- Decimal scaling: $j = 2$, not 1.** $j = 1$ would give $70/10 = 7 \not< 1$. Check the *maximum*, not the value being transformed.
- All three land near 0.35–0.39 here, which is coincidence: 35 sits just above the mean and just under the middle of the range.

Step by step: bins of three, then the value 35 three ways



M2. PCA: principal components and proportion of variance

Method

1. Start from the **covariance matrix** Σ (if given raw data: subtract the mean first, then compute Σ).
2. Solve the characteristic equation $|\Sigma - \lambda I| = 0$ for the eigenvalues λ .
3. Sort λ in **descending** order. λ_i = variance carried by component i .
4. For each λ_i solve $(\Sigma - \lambda_i I) \mathbf{v} = 0$ for the eigenvector, then normalise it to unit length.
5. Proportion of total variance explained by component i :

$$\frac{\lambda_i}{\sum_j \lambda_j}, \quad \sum_i \lambda_i = \text{trace}(\Sigma)$$

- **Check:** the eigenvalues must sum to the trace. Free and catches arithmetic slips.

Problem 2.1 4

QUESTION AS PRINTED 2076 Chaitra · Q1 · [4]

Find the principal components and the proportion of the total variance explained by each when the covariance matrix of the three random variables X_1 , X_2 and X_3 is:

$$\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Step 1: characteristic equation

- The third row/column is decoupled, so Σ splits into a 2×2 block and a 1×1 block.
- From the 1×1 block: $(2 - \lambda) = 0 \Rightarrow \boxed{\lambda = 2}$
- From the 2×2 block:

$$(1 - \lambda)(5 - \lambda) - (-2)(-2) = 0$$

$$5 - 6\lambda + \lambda^2 - 4 = 0 \Rightarrow \lambda^2 - 6\lambda + 1 = 0$$

$$\lambda = \frac{6 \pm \sqrt{36 - 4}}{2} = \frac{6 \pm \sqrt{32}}{2} = 3 \pm 2\sqrt{2}$$

Step 2: eigenvalues, sorted

- $\lambda_1 = 3 + 2\sqrt{2} = \mathbf{5.8284}$
- $\lambda_2 = \mathbf{2}$
- $\lambda_3 = 3 - 2\sqrt{2} = \mathbf{0.1716}$
- **Check:** $5.8284 + 2 + 0.1716 = 8 = \text{trace}(\Sigma) = 1 + 5 + 2$. **Correct.**

Step 3: eigenvectors

- For $\lambda_1 = 5.8284$: row 1 gives $(1 - 5.8284)v_1 - 2v_2 = 0 \Rightarrow v_2 = -2.4142 v_1$; row 3 gives $v_3 = 0$.
 $\mathbf{e}_1 \propto (1, -2.4142, 0)$, length $\sqrt{1 + 5.8284} = 2.6131$
 $\mathbf{e}_1 = (0.3827, -0.9239, 0)$
- For $\lambda_2 = 2$: rows 1 and 2 force $v_1 = v_2 = 0$, v_3 free.
 $\mathbf{e}_2 = (0, 0, 1)$
- For $\lambda_3 = 0.1716$: row 1 gives $0.8284 v_1 = 2v_2 \Rightarrow v_2 = 0.4142 v_1$; $v_3 = 0$.
 $\mathbf{e}_3 \propto (1, 0.4142, 0)$, length 1.0824
 $\mathbf{e}_3 = (0.9239, 0.3827, 0)$
- Check:** $\mathbf{e}_1 \cdot \mathbf{e}_3 = (0.3827)(0.9239) + (-0.9239)(0.3827) = 0$
 $\rightarrow$ orthogonal, as required for a symmetric matrix.

Final answer

- PC1: $\lambda_1 = 5.8284$, $\mathbf{e}_1 = (0.3827, -0.9239, 0)$, **72.86%**
- PC2: $\lambda_2 = 2$, $\mathbf{e}_2 = (0, 0, 1)$, **25.00%**
- PC3: $\lambda_3 = 0.1716$, $\mathbf{e}_3 = (0.9239, 0.3827, 0)$, **2.14%**

Step by step: eigenvalues, the check, the vectors, the variance kept

1. Spot the blocks

block split

1	-2	0
-2	5	0
0	0	2

2×2 block and a 1×1 block: solve each alone.

1×1 : $2 - \lambda = 0 \Rightarrow \lambda = 2$

2. Eigenvalues of the 2×2

characteristic eq.

$$(1 - \lambda)(5 - \lambda) - 4 = 0$$

$$\lambda^2 - 6\lambda + 1 = 0$$

$$\lambda = 3 \pm 2\sqrt{2}$$

$\lambda_1 = 5.8284$, $\lambda_2 = 2$, $\lambda_3 = 0.1716$

KEY STEP

3. Trace check

trace check

5.83	2
------	---

$5.8284 + 2 + 0.1716 = 8$
 trace = $1 + 5 + 2 = 8$ agrees

4. Eigenvectors

eigenvector

$\mathbf{e}_1 = (0.3827, -0.9239, 0)$
 $\mathbf{e}_3 = (0.9239, 0.3827, 0)$
 $\mathbf{e}_2 = (0, 0, 1)$, out of the page
 $\mathbf{e}_1 \cdot \mathbf{e}_3 = 0$: at right angles

5. Share of variance

share of variance

72.86%	25.00%
--------	--------

PC1 + PC2 = **97.86%**; PC3 only 2.14%.
 Keep 2 components: 3 dims $\rightarrow$ 2.

M3. Distance measures and the distance matrix**Method**

- All three are Minkowski with a different r :

$$\text{dist}(p, q) = \left(\sum_{k=1}^n |p_k - q_k|^r \right)^{1/r}$$

- $r = 1$ **Manhattan**: $\sum |p_k - q_k|$
- $r = 2$ **Euclidean**: $\sqrt{\sum (p_k - q_k)^2}$
- $r \rightarrow \infty$ **Supremum** (L_∞ , Chebyshev): $\max_k |p_k - q_k|$
- A **distance matrix** is symmetric with a zero diagonal, so only the lower triangle needs computing.

Warm-up (Han & Kamber Example 2.19–2.20)

PROBLEM not a PYQ · Han & Kamber Example 2.19–2.20 verified/added

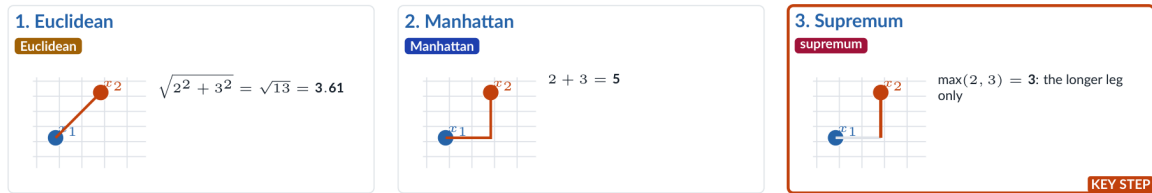
Given the two objects $x_1 = (1, 2)$ and $x_2 = (3, 5)$, compute the Euclidean, Manhattan and Supremum distances between them.

Smallest case that shows all three at once. Worth memorising as a check.

- $x_1 = (1, 2)$, $x_2 = (3, 5)$. Differences: 2 and 3.
- Euclidean = $\sqrt{2^2 + 3^2} = \sqrt{13} = \mathbf{3.61}$

- Manhattan = $2 + 3 = 5$
- Supremum = $\max(2, 3) = 3$
- **Ordering**: $\text{supremum} \leq \text{Euclidean} \leq \text{Manhattan}$. Always true. Use it to catch slips.

Step by step: the same two points, three distances



Problem 3.1 2

QUESTION AS PRINTED 2076 Chaitra · Q2 (a) · [2]

Given the following points compute the distance matrix using the Manhattan and the Supremum distance.

Points	X	Y
P1	6	3
P2	2	2
P3	3	4

Manhattan (L_1)

- $d(P_1, P_2) = |6 - 2| + |3 - 2| = 4 + 1 = 5$
- $d(P_1, P_3) = |6 - 3| + |3 - 4| = 3 + 1 = 4$
- $d(P_2, P_3) = |2 - 3| + |2 - 4| = 1 + 2 = 3$

Supremum (L_∞)

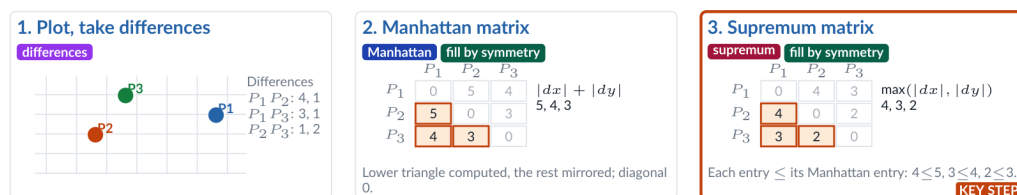
- $d(P_1, P_2) = \max(4, 1) = 4$
- $d(P_1, P_3) = \max(3, 1) = 3$
- $d(P_2, P_3) = \max(1, 2) = 2$

	P_1	P_2	P_3
P_1	0	5	4
P_2	5	0	3
P_3	4	3	0

	P_1	P_2	P_3
P_1	0	4	3
P_2	4	0	2
P_3	3	2	0

- **Sanity check**: $\text{supremum} \leq \text{Euclidean} \leq \text{Manhattan}$, always. Holds here ($4 \leq 5$, $3 \leq 4$, $2 \leq 3$).

Step by step: plot, then fill each matrix's lower triangle



Problem 3.2 3

QUESTION AS PRINTED 2080 Bhadra · Q1 (b) · [3+3+3]

How do similarity / dissimilarity is calculated? Find the cosine similarity between Object-2 and 4. Also calculate the Euclidean distance between object 1, 3 and object 1, 4.

Object	Size	Weight	Color_Code	Taste_Score
1	4	56	7	10
2	3	53	8	11
3	7	58	6	9
4	9	55	7	12

The cosine-similarity part of this same question is Problem 4.1 below.

Step-by-step

- $d(O_1, O_3)$: differences $(4 - 7, 56 - 58, 7 - 6, 10 - 9) = (-3, -2, 1, 1)$
 $= \sqrt{9 + 4 + 1 + 1} = \sqrt{15} = \mathbf{3.873}$
- $d(O_1, O_4)$: differences $(4 - 9, 56 - 55, 7 - 7, 10 - 12) = (-5, 1, 0, -2)$
 $= \sqrt{25 + 1 + 0 + 4} = \sqrt{30} = \mathbf{5.477}$

Final answer

$d(O_1, O_3) = 3.873$, $d(O_1, O_4) = 5.477$. Object 1 is closer to object 3 than to object 4.

Step by step: differences, then square, sum and root

1. Differences

differences

$O_1 - O_3$

Size	Wt	Col	Taste
-3	-2	1	1

$O_1 - O_4$

Size	Wt	Col	Taste
-5	1	0	-2

2. Square, sum, root

Euclidean

$d(O_1, O_3) = \sqrt{9 + 4 + 1 + 1} = \sqrt{15} = \mathbf{3.873}$

$d(O_1, O_4) = \sqrt{25 + 1 + 0 + 4} = \sqrt{30} = \mathbf{5.477}$

Object 1 is closer to 3 than to 4.

Square, add, root: the size gap of 5 dominates $d(O_1, O_4)$.

KEY STEP

M4. Cosine similarity**Method**

$$\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{\|d_1\| \|d_2\|}, \quad d_1 \cdot d_2 = \sum_k d_{1k} d_{2k}, \quad \|d\| = \sqrt{\sum_k d_k^2}$$

- Measures **angle only**, ignores magnitude → used for documents.
- Always lay out 3 lines: dot product, then each length, then divide.

Typo in your class notes verified/added : DBK's Lecture-2 cosine example prints $\|d_2\| = (6)^{0.5} = 2.245$. In fact $\sqrt{6} = \mathbf{2.449}$. His final answer 0.3150 is correct and only follows from $2.449 (5 / (6.481 \times 2.449) = 0.315)$, so 2.245 is a slip. Copy the method, not that number.

Problem 4.1 3

QUESTION AS PRINTED 2080 Bhadra · Q1 (b) · [3+3+3]

How do similarity / dissimilarity is calculated? Find the cosine similarity between Object-2 and 4. Also calculate the Euclidean distance between object 1, 3 and object 1, 4.

Object	Size	Weight	Color_Code	Taste_Score
1	4	56	7	10
2	3	53	8	11
3	7	58	6	9
4	9	55	7	12

The Euclidean-distance part of this same question is Problem 3.2 above.

Given

$O_2 = (3, 53, 8, 11)$, $O_4 = (9, 55, 7, 12)$.

Step-by-step

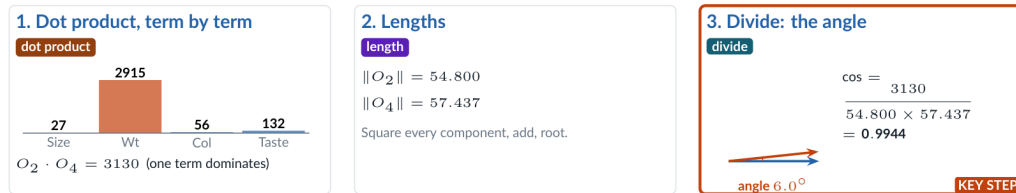
- Dot product: $3(9) + 53(55) + 8(7) + 11(12) = 27 + 2915 + 56 + 132 = 3130$
- $\|O_2\| = \sqrt{9 + 2809 + 64 + 121} = \sqrt{3003} = 54.800$
- $\|O_4\| = \sqrt{81 + 3025 + 49 + 144} = \sqrt{3299} = 57.437$
- $\cos = \frac{3130}{54.800 \times 57.437} = \frac{3130}{3147.5}$

Final answer

$\cos(O_2, O_4) = \mathbf{0.9944}$.

- Nearly 1 → the two objects point almost the same direction.
- **Why so high:** *Weight* (≈ 50) dwarfs the other attributes and dominates the dot product. **Normalise before using cosine** on mixed-scale attributes, or the answer means very little. verified/added

Step by step: cosine of objects 2 and 4 (the weight term dominates)



Problem 4.2 1

QUESTION AS PRINTED 2080 Baishakh · Q2 (b)(iv) · [1]

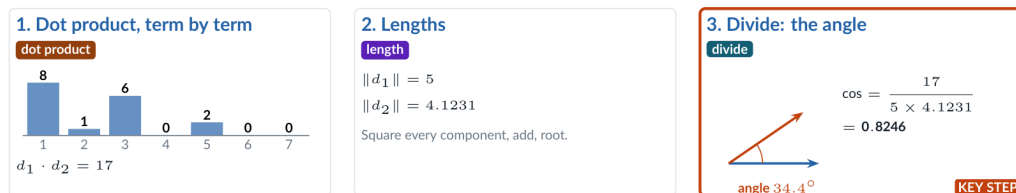
Find the Cosine similarity between documents $d_1 = (4, 1, 2, 0, 2, 0, 0)$ and $d_2 = (2, 1, 3, 0, 1, 1, 1)$. (the rest of Q2 b is Problem 5.2 below)

- Dot: $8 + 1 + 6 + 0 + 2 + 0 + 0 = 17$
- $\|d_1\| = \sqrt{16 + 1 + 4 + 0 + 4 + 0 + 0} = \sqrt{25} = 5$
- $\|d_2\| = \sqrt{4 + 1 + 9 + 0 + 1 + 1 + 1} = \sqrt{17} = 4.1231$
- $\cos = 17 / (5 \times 4.1231) = 17 / 20.6155$

Final answer

0.8246

Step by step: cosine of the two documents



Problem 4.3 1

QUESTION AS PRINTED 2076 Chaitra · Q2 (b) · [1]

Given the following two vectors compute the Cosine similarity between them.

$$D_1 = [4 \ 0 \ 2 \ 0 \ 1] \quad D_2 = [2 \ 0 \ 0 \ 2 \ 2]$$

- Dot: $8 + 0 + 0 + 0 + 2 = 10$
- $\|D_1\| = \sqrt{16 + 0 + 4 + 0 + 1} = \sqrt{21} = 4.5826$
- $\|D_2\| = \sqrt{4 + 0 + 0 + 4 + 4} = \sqrt{12} = 3.4641$
- $\cos = 10 / (4.5826 \times 3.4641) = 10 / 15.8745$

Final answer

0.6299

Step by step: cosine of the two vectors

1. Dot product, term by term

dot product



$D_1 \cdot D_2 = 10$ (one term dominates)

2. Lengths

length

$\|D_1\| = 4.5826$
 $\|D_2\| = 3.4641$

Square every component, add, root.

3. Divide: the angle

divide

$\cos = \frac{10}{4.5826 \times 3.4641} = 0.6299$



angle 51.0° **KEY STEP**

M5. Binary similarity: Jaccard, SMC, symmetric dissimilarity

Method

- Encode each object as a 0/1 vector, then count over all attribute positions:
 - M_{11} : both 1 M_{00} : both 0
 - M_{10} : $p = 1, q = 0$ M_{01} : $p = 0, q = 1$
- SMC** (symmetric binary, 0-0 counts): $\frac{M_{11} + M_{00}}{M_{11} + M_{00} + M_{10} + M_{01}}$
- Jaccard** (asymmetric binary, 0-0 ignored): $\frac{M_{11}}{M_{11} + M_{10} + M_{01}}$
- Symmetric binary dissimilarity**: $d = \frac{M_{10} + M_{01}}{M_{11} + M_{00} + M_{10} + M_{01}} = 1 - SMC$
- Nominal (non-binary) dissimilarity**: $d = (p - m)/p$, m = matches, p = total attrs.

Problem 5.1 2

QUESTION AS PRINTED 2076 Chaitra · Q2 (c) · [2]

Given the following two binary vectors compute the Jaccard similarity and Simple Matching Coefficient.

$$P = [0 \ 0 \ 1 \ 1 \ 0 \ 1] \quad Q = [1 \ 1 \ 1 \ 1 \ 0 \ 1]$$

Step-by-step: tabulate position by position

Position	1	2	3	4	5	6
P	0	0	1	1	0	1
Q	1	1	1	1	0	1
Category	M_{01}	M_{01}	M_{11}	M_{11}	M_{00}	M_{11}

- $M_{11} = 3, M_{01} = 2, M_{10} = 0, M_{00} = 1$ (total = 6 ✓)
- $J = \frac{3}{3 + 0 + 2} = \frac{3}{5}$
- $SMC = \frac{3 + 1}{6} = \frac{4}{6}$

Final answer

 $J = 0.6$, $SMC = 0.667$. SMC is higher because it credits the one 0-0 match that Jaccard discards.

Step by step: tally the pairs, then Jaccard and SMC

1. Tally position by position

tally the pairs

P	0	0	1	1	0	1
Q	1	1	1	1	0	1
pair	01	01	11	11	00	11

$M_{11}=3 \ M_{10}=0 \ M_{01}=2 \ M_{00}=1$

KEY STEP

2. Jaccard

Jaccard

$$J = \frac{M_{11}}{M_{11} + M_{10} + M_{01}} = \frac{3}{5} = 0.6$$

The single 0-0 (position 5) is left out.

3. SMC

SMC

$$SMC = \frac{M_{11} + M_{00}}{6} = \frac{4}{6} = 0.667$$

The 0-0 counts as agreement, so $SMC > J$.

Problem 5.2 2+2+2

QUESTION AS PRINTED 2080 Baishakh · Q2 (b) · [2+2+2+1]

Consider the following table:

Name	Gender	Eyecolor	Haircolor	Test-1	Test-2	Fever	Cough
Ram	M	Black	Gray	P	N	P	N
Laxmi	F	Blue	Black	P	P	N	N
Shyam	M	Blue	Gray	N	P	N	P

- Calculated Jaccard Coefficient of $Sim_{jaccard}$ (Ram, Laxmi) for asymmetric binary attributes.
- Dissimilarity of symmetric binary attributes d (Laxmi, Shyam).
- Find the Simple matching coefficients of SMC (Ram, Shyam).
- Find the Cosine similarity between documents $d_1 = (4, 1, 2, 0, 2, 0, 0)$ and $d_2 = (2, 1, 3, 0, 1, 1, 1)$ — solved as Problem 4.2 above.

Step 0: split the attributes (the whole question turns on this)

- Asymmetric binary** = Test-1, Test-2, Fever, Cough. Encode $P = 1$, $N = 0$. A shared “N” is uninformative, which is exactly what “asymmetric” means.
- Symmetric** = Gender, Eyecolor, Haircolor. Both/all states are equally meaningful.
- Encoded: Ram = (1, 0, 1, 0), Laxmi = (1, 1, 0, 0), Shyam = (0, 1, 0, 1).

(i) Jaccard(Ram, Laxmi)

	T1	T2	Fev	Cou
Ram	1	0	1	0
Laxmi	1	1	0	0
Cat.	M_{11}	M_{01}	M_{10}	M_{00}

$$- M_{11} = 1, M_{01} = 1, M_{10} = 1, M_{00} = 1$$

$$- J = \frac{1}{1 + 1 + 1} = \frac{1}{3}$$

Answer

$$J = 0.333$$

(iii) SMC(Ram, Shyam)

	T1	T2	Fev	Cou
Ram	1	0	1	0
Shyam	0	1	0	1
Cat.	M_{10}	M_{01}	M_{10}	M_{01}

$$- M_{11} = 0, M_{00} = 0, M_{10} = 2, M_{01} = 2$$

$$- SMC = \frac{0 + 0}{4} = 0$$

Answer

$SMC = 0$ (they disagree on every test).

(ii) d (Laxmi, Shyam), symmetric binary attributes

- Across these 3 people each of Gender / Eyecolor / Haircolor takes exactly **2 states**, and neither state is an “absence” → all 3 are **symmetric binary**.
- Encode: Gender M=1 · Eyecolor Black=1 · Haircolor Gray=1.

	Gender	Eye	Hair
Laxmi (F, Blue, Black)	0	0	0
Shyam (M, Blue, Gray)	1	0	1
Category	s	t	s

$$- q = 0, r = 0, s = 2, t = 1 \quad (\text{total } p = 3 \checkmark)$$

- Symmetric binary dissimilarity keeps t :

$$d = \frac{r + s}{q + r + s + t} = \frac{0 + 2}{3} = \frac{2}{3}$$

Answer

$$d = 0.667$$

Cross-check: for binary attrs the symmetric formula reduces to the nominal mismatch ratio $(p - m)/p = (3 - 1)/3 = 2/3$. Same answer by both routes, so the encoding choice is safe.

verified/added

Step by step: decide the attribute types first, then tally each pair

1. Split the attributes

split attributes

symmetric

Gender, Eye, Hair
both states count

asymmetric

Test-1, Test-2,
Fever, Cough
shared N = no info

P → 1, N → 0 on the tests.

KEY STEP

2. (i) Jaccard(Ram, Laxmi)

tally the pairs Jaccard

	T1	T2	Fev	Cou
Ram	1	0	1	0
Laxmi	1	1	0	0

pair 11 01 10 00

$M_{11}=1 \quad M_{10}=1 \quad M_{01}=1 \quad M_{00}=1$

$J = 1/(1 + 1 + 1) = 0.333$

3. (ii) d (Laxmi, Shyam), symmetric

tally the pairs symmetric d

	Gen	Eye	Hair
Laxmi	0	0	0
Shyam	1	0	1

pair 01 00 01

$M_{11}=0 \quad M_{10}=0 \quad M_{01}=2 \quad M_{00}=1$

$d = (0 + 2)/3 = 0.667$

4. (iii) SMC(Ram, Shyam)

tally the pairs SMC

	T1	T2	Fev	Cou
Ram	1	0	1	0
Shyam	0	1	0	1

pair 10 01 10 01

$M_{11}=0 \quad M_{10}=2 \quad M_{01}=2 \quad M_{00}=0$

$SMC = (0 + 0)/4 = 0$

M6. Schema design + OLAP operation sequence

Method

1. Put the **measures** in a central **fact table**, plus one foreign key per dimension.
2. Draw one **dimension table** per dimension around it.
 - **Star** → leave dimension tables flat (de-normalised).
 - **Snowflake** → normalise the hierarchies into sub-tables (location → city → province).
3. For the OLAP sequence, work from the base cuboid and ask, per dimension:
 - Need a coarser level? → **roll-up**
 - Need a finer level? → **drill-down**
 - Restricting 1 dimension? → **slice**. Restricting ≥ 2 ? → **dice**
 - Dimension not wanted at all? → **roll-up to all** (dimension reduction)
4. n dimensions → 2^n cuboids in the lattice.

Problem 6.1 3+3

QUESTION AS PRINTED 2078 Bhadra · Q2 · [3+3]

Suppose that a data warehouse consists of the four dimensions data, spectator, location, and game, and the two measures count and charge, where charge is the fare that a spectator pays when watching a game on a given date. Spectators may be students, adults or seniors, with each category having its own charge rate.

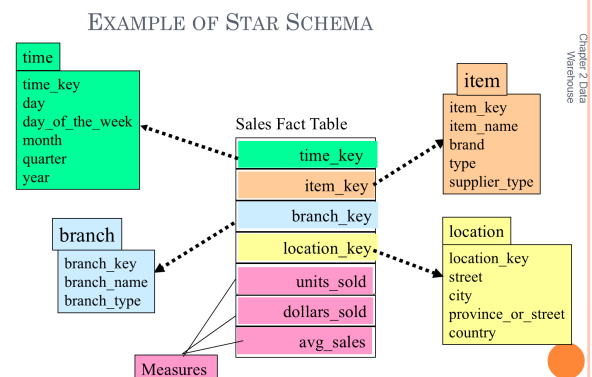
- a) Draw a star schema diagram for the data warehouse.
- b) Starting with the base cuboid [data, spectator, location, game], what specific OLAP operations should you perform in order to list the total charge paid by student spectators at Dashrath Stadium in 2021?

(a) Star schema

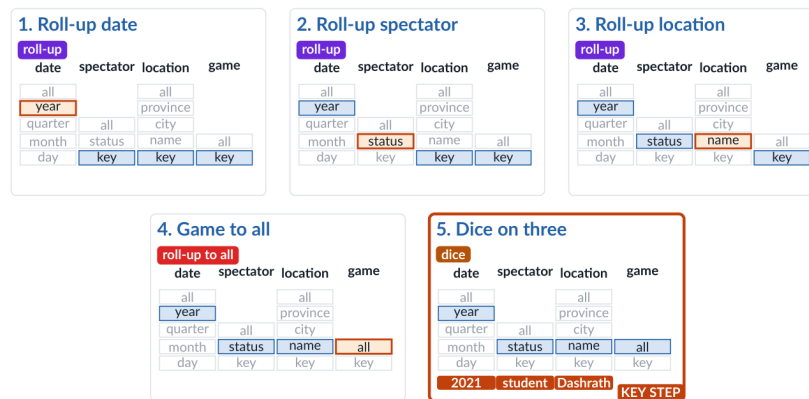
- **Fact table** Sales: date_key, spectator_key, location_key, game_key, count, charge.
- **Dimension tables** (flat, one hop from the fact table):
 - date(date_key, day, month, quarter, year)
 - spectator(spectator_key, name, status, phone, address) where status $\in$ {student, adult, senior}
 - location(location_key, name, street, city, province, country)
 - game(game_key, name, description, sport_type)
- 4 dimensions → $2^4 = 16$ cuboids in the lattice.

(b) OLAP operation sequence

1. **Roll-up** on *date* from date_key to year.
 2. **Roll-up** on *spectator* from spectator_key to status.
 3. **Roll-up** on *location* from location_key to location_name.
 4. **Roll-up** on *game* to all (the question does not split by game).
 5. **Dice** with year = 2021 and status = student and location_name = ``Dashrath Stadium''.
 6. Read off the charge measure (summed).
- **Dice, not slice** → 3 dimensions are being restricted at once.



Step by step: each dimension's level after every OLAP operation (orange = changed)



Problem 6.2 3+3

QUESTION AS PRINTED 2076 Chaitra · Q3 · [3+3]

Suppose that a data warehouse for a sales company consists of five dimensions: *time*, *location*, *supplier*, *brand*, and *product*, and two measures: *count* and *price*.

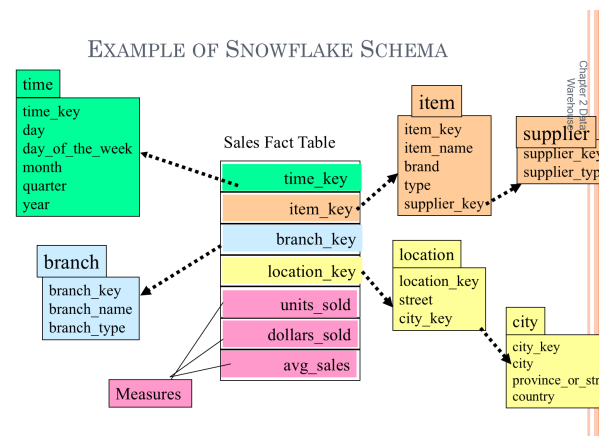
- Draw a *snowflake schema* diagram for the data warehouse.
- Starting with the base cuboid [*time*, *location*, *supplier*, *brand*, *product*], what specific OLAP operations should one perform in order to list the total count for a certain *brand* for each *state* per *year* (assume *location* has three levels: *country*, *state*, *city*; and assume *time* has three levels: *year*, *month*, *day*)?

(a) Snowflake schema

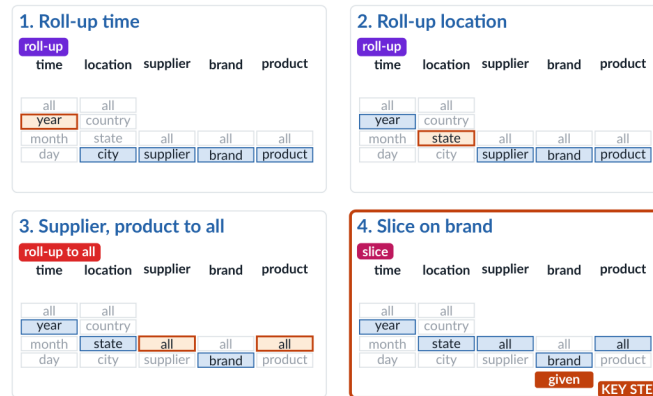
- Fact table** Sales: *time_key*, *location_key*, *supplier_key*, *brand_key*, *product_key*, *count*, *price*.
- Normalised hierarchies** (this is what makes it a snowflake, not a star):
 - time*(*time_key*, *day*, *month_key*) → *month*(*month_key*, *month*, *year*)
 - location*(*location_key*, *city_key*) → *city*(*city_key*, *city*, *state_key*) → *state*(*state_key*, *state*, *country*)
 - product*(*product_key*, *name*, *brand_key*) → *brand*(*brand_key*, *brand_name*)
 - supplier*(*supplier_key*, *supplier_name*, *supplier_type*)
- 5 dimensions → $2^5 = 32$ cuboids.

(b) OLAP operation sequence

- Roll-up on *time* from *day* to *year*.
 - Roll-up on *location* from *city* to *state*.
 - Roll-up on *supplier* to all (not wanted).
 - Roll-up on *product* to all: *brand* is its own dimension of this cube, so the *product* dimension is not needed.
 - Slice on *brand* = the given brand (1 dimension restricted).
 - Result: cuboid [*year*, *state*] holding total count.
- Do not roll product up to its brand level** as well: *brand* would then be held twice, once as a dimension and once as a level of *product*.



Step by step: each dimension's level after every OLAP operation (orange = changed)



M7. Central tendency, quartiles, IQR and outliers

NEW Not in any of the 23 papers. Added because Notes by PC/unit 2 Numerical.docx and Untitled spreadsheet.xlsx (2025-26) are built entirely around it. Theory in Ch 2, §2.5.

Method

- **Grouped median** = $L_1 + \left(\frac{N/2 - (\sum freq)_i}{freq_{median}} \right) \times width$. Find the median interval first: the one where cumulative frequency first reaches $N/2$.
- **Mode from mean and median** $\approx 3 \text{ median} - 2 \text{ mean}$ (unimodal, moderately skewed).
- **Quartiles**, interpolated convention: position of $Q_k = \frac{k(n+1)}{4}$, then interpolate between the neighbouring sorted values.
- **Outlier fences**: below $Q_1 - 1.5 IQR$ or above $Q_3 + 1.5 IQR$.

Problem 7.1 – grouped median, 100 employees 4

PROBLEM not a PYQ · PC's notes · [4]

Suppose the salaries of 100 employees of a company are grouped into \$10,000 intervals as shown in the table below. Find the median salary.

Salary (\$)	Freq	Cum freq
10,000–20,000	8	8
20,000–30,000	15	23
30,000–40,000	20	43
40,000–50,000	32	75
50,000–60,000	18	93
60,000–70,000	7	100

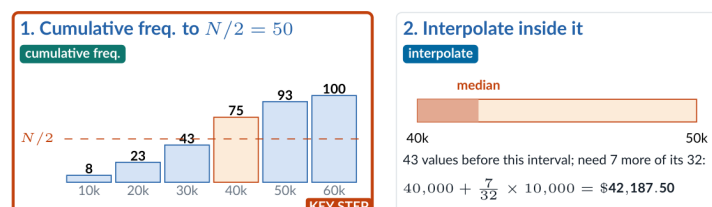
Solution

1. $N = 100 \rightarrow N/2 = 50$. Need the 50th observation.
2. Cumulative frequency passes 50 inside **40k–50k** (43 before it, 75 after) $\rightarrow$ that is the median interval.
3. $L_1 = 40,000$, $(\sum freq)_i = 43$,
 $freq_{median} = 75 - 43 = 32$, $width = 10,000$.
4. $median = 40,000 + \frac{50-43}{32} \times 10,000 = 40,000 + \frac{7}{32} \times 10,000$

Answer

$median = \$42,187.50$

Step by step: find the median interval, then interpolate



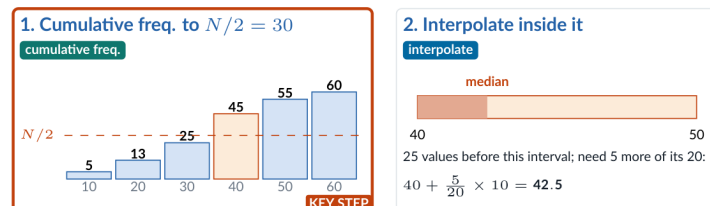
Problem 7.2 – same method, smaller table 3

PROBLEM not a PYQ · PC's notes · [3]

Salaries (in \$1000s) of 60 employees fall in the intervals 10–20, 20–30, 30–40, 40–50, 50–60 and 60–70 with frequencies 5, 8, 12, 20, 10 and 5 respectively. Find the median salary.

- $N = 5 + 8 + 12 + 20 + 10 + 5 = 60 \rightarrow N/2 = 30$.
- Cumulative before 40–50 = $5 + 8 + 12 = 25$; $freq_{median} = 20$, $L_1 = 40$, $width = 10$.
- $median = 40 + \frac{30-25}{20} \times 10 = 40 + 2.5 = 42.5$ (i.e. \$42,500).

Step by step: find the median interval, then interpolate



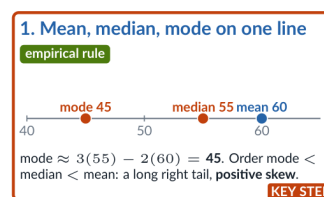
Problem 7.3 – mode from mean and median 2

PROBLEM not a PYQ · PC's notes · [2]

For a unimodal, moderately skewed distribution the mean is 60 and the median is 55. Estimate the mode, and state what the result says about the skew.

- $mean - mode \approx 3(mean - median) \Rightarrow mode \approx 3median - 2mean$.
- $mode \approx 3(55) - 2(60) = 165 - 120 = 45$.
- **Sanity check:** $mode < median < mean \rightarrow$ positively skewed. Consistent.

Step by step: the three centres on one line



Problem 7.4 – full descriptive summary of one dataset 2+2+2+2

PROBLEM not a PYQ · Han & Kamber Example 2.6 · [2+2+2+2]

Suppose that the data for analysis is the salaries (in \$1000s) of 12 employees, already sorted in increasing order:

30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110

Find (a) the mean, median and mode; (b) the quartiles Q_1 and Q_3 and the IQR; (c) the five-number summary; (d) any outliers using the $1.5 \times IQR$ rule.

(a) Central tendency

- $\sum x = 696 \rightarrow \text{mean} = \frac{696}{12} = 58.$
- $n \text{ even} \rightarrow \text{median} = \frac{52+56}{2} = 54.$
- 52 and 70 each occur twice $\rightarrow$ **bimodal**, modes **52** and **70**.
- $\text{midrange} = \frac{30+110}{2} = 70.$
- $\text{mode} < \text{median} < \text{mean}$ is broken here by the bimodality; the single outlier 110 is what pulls the mean above the median.

(b) Quartiles, positional convention (Han & Kamber)

- Take the 3rd, 6th, 9th sorted values $\rightarrow Q_1 = 47, Q_2 = 52, Q_3 = 63.$
- $IQR = 63 - 47 = 16.$

(c) Quartiles, interpolated convention (PC's class notes)

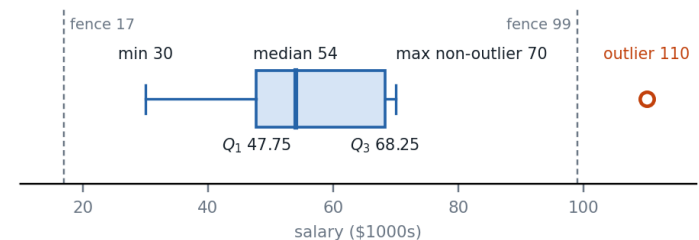
- $Q_1 \text{ at } \frac{1(13)}{4} = 3.25 \rightarrow 47 + 0.25(50 - 47) = 47.75$
- $Q_2 \text{ at } \frac{2(13)}{4} = 6.5 \rightarrow 52 + 0.5(56 - 52) = 54$
- $Q_3 \text{ at } \frac{3(13)}{4} = 9.75 \rightarrow 63 + 0.75(70 - 63) = 68.25$
- $IQR = 68.25 - 47.75 = 20.5$

(d) Five-number summary and outliers

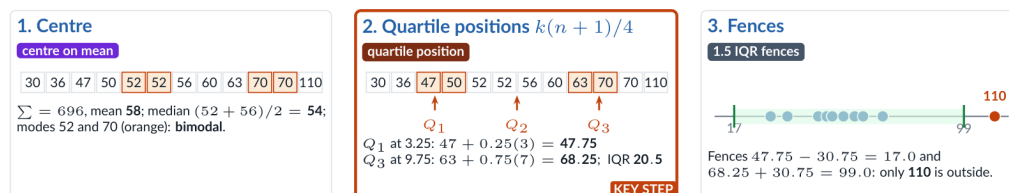
- Five-number summary (interpolated): **30, 47.75, 54, 68.25, 110.**
- Fences: $47.75 - 1.5(20.5) = 17.0$ and $68.25 + 1.5(20.5) = 99.0.$
- $110 > 99 \rightarrow$ **110** is a suspected outlier. Nothing below 17.
- **Cross-check with the other convention:** fences $47 - 24 = 23$ and $63 + 24 = 87.$ **110 is still the only outlier** $\rightarrow$ the verdict does not depend on the convention, so either is safe in the exam as long as you name it.

(e) Boxplot

- Box from Q_1 to Q_3 , line at 54, whiskers to 30 and to the largest non-outlier (70), and **110 plotted as a lone point.**



Drawn to scale by `boxfig.py` from the values above. The whisker stops at 70, not at the fence.

Step by step: centre, quartiles, fences**Extra practice (patterns not yet fully covered)****Practice A – binning for noise smoothing**

Binning is examined in prose 72 Ka, 76 Ash but never as a numerical. **This is DBK's own worked example** (Lecture-1, "Binning Methods for Data Smoothing"), so it is the likeliest form any future numerical would take. Answers below independently recomputed and they match his slide exactly.

PROBLEM not a PYQ · DBK Lecture-1 worked example **verified/added**

The following price data has already been sorted:

4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

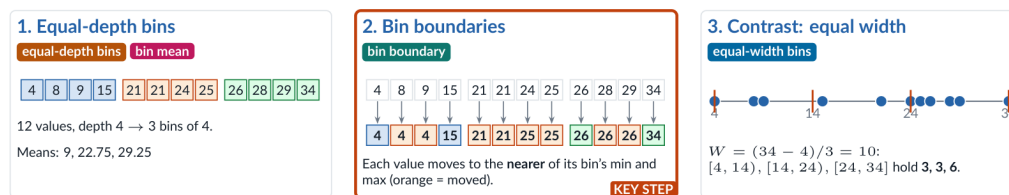
Partition the data into **equal-depth** bins of depth 4, then smooth the noise by (i) bin means and (ii) bin boundaries.

Solution

- 12 values, depth 4 $\rightarrow$ 3 bins.

- Bin 1: 4, 8, 9, 15 Bin 2: 21, 21, 24, 25 Bin 3: 26, 28, 29, 34
- (i) **Smoothing by bin means**
 - Bin 1 mean = $36/4 = 9 \rightarrow 9, 9, 9, 9$
 - Bin 2 mean = $91/4 = 22.75 \approx 23 \rightarrow 23, 23, 23, 23$
 - Bin 3 mean = $117/4 = 29.25 \approx 29 \rightarrow 29, 29, 29, 29$
- (ii) **Smoothing by bin boundaries** (each value $\rightarrow$ nearer of min/max)
 - Bin 1 (min 4, max 15): 4, 8, 9, 15 $\rightarrow 4, 4, 4, 15$ (8 and 9 are nearer 4)
 - Bin 2 (min 21, max 25): 21, 21, 24, 25 $\rightarrow 21, 21, 25, 25$
 - Bin 3 (min 26, max 34): 26, 28, 29, 34 $\rightarrow 26, 26, 26, 34$
- **Equal-width for contrast:** range = $34 - 4 = 30$, $N = 3 \rightarrow W = (B - A)/N = 10$. Intervals $[4, 14)$, $[14, 24)$, $[24, 34]$ $\rightarrow$ counts 3, 3, 6. **Unequal counts: that is the difference from equal-depth.**

Step by step: equal-depth bins, boundaries, and equal width for contrast



Practice B — min-max into a range other than [0,1]

Every PYQ so far uses $[0, 1]$. The figures below are Han & Kamber 3rd ed Example 3.4, so they are the canonical ones. **verified/added**

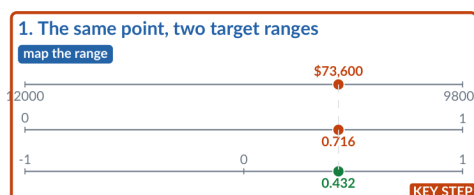
PROBLEM not a PYQ · Han & Kamber 3rd ed Example 3.4 **verified/added**

Suppose that the minimum and maximum values for the attribute *income* are \$12,000 and \$98,000 respectively. Normalise **\$73,600** by min-max normalization, first to the range $[0, 1]$ and then to the range $[-1, 1]$.

Solution

- To $[0, 1]$: $v' = \frac{73600 - 12000}{98000 - 12000}(1 - 0) + 0 = \frac{61600}{86000} = \mathbf{0.716}$
- To $[-1, 1]$: $v' = \frac{61600}{86000}(1 - (-1)) + (-1) = 0.716(2) - 1 = \mathbf{0.432}$
- **Check:** the value sits 71.6% of the way up the range, so in $[-1, 1]$ it must land slightly above the midpoint 0. It does.

Step by step: one income, mapped into two ranges



Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity-relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /23** = topic asked in 7 or more of the 17 papers · **HOT /23** = 4–6 papers · **PIN /23** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.